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Supplementary Materials for

Estimating human joint moments unifies exoskeleton control, reducing user effort

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Data files S1 and S2
MDAR Reproducibility Checklist

Supplementary Methods

The Supplementary Methods provide additional details and information pertaining to our study. The first several sections follow a similar outline compared to the Materials and Methods in the Main Text. First, a detailed explanation of the experimental methods used in each phase of the study protocol is provided. Second, additional information describing the robotic hip exoskeleton hardware, electronics, and software is presented. Third, the hyperparameters and supervised training methods of the temporal convolutional network (TCN) are discussed. Fourth, the biomechanical modeling methods used to compute ground-truth joint kinematics and moments are provided. Finally, detailed descriptions of the methods used to compute metabolic cost and lower-limb joint work are presented.

Additionally, this document contains four supplementary analyses. The first describes our methods and results for transforming the exoskeleton sensor data across devices. The second investigates the effects of mid-level controller delay magnitude on user metabolic cost. The third analyzes hip moment estimation performance when tested on ambulatory conditions that were not in the training set. Finally, the fourth analysis quantifies the effect of the model training set size on hip moment estimation accuracy.

Detailed Description of the Experimental Protocol

As mentioned in the Materials and Methods, the experimental protocol was conducted over four phases, in which all participants provided informed consent according to the protocols approved by the Georgia Institute of Technology Institutional Review Board. Phase 1 consent was obtained under the approved protocol H17218. Phase 2 consent was obtained under the approved protocols H20533 and H22051. Consent for Phases 3 and 4 was obtained under the approved protocol H22051. The methods used for data collection and the details of each phase of the protocol are described in the following sections.

Experimental Data Collection

For trials that included biomechanical measurements used to compute ground-truth joint kinematics and moments, motion capture data were collected at 200 Hz (Vicon Motion Systems, United Kingdom), and ground reaction force (GRF) data were collected at 1000 Hz (Bertec, OH, USA). GRF data were synced natively with the motion capture data using a Vicon Lock Sync Box. For all metabolic trials, volumetric flowrates of oxygen intake ($\dot{V}O_2$) and carbon dioxide exhaust ($\dot{V}CO_2$) from the breaths of each participant were measured using a metabolic measurement system (TrueOne 2400, ParvoMedics, UT, USA) and used to compute metabolic cost (further details in Analyzing Exoskeleton Effects on User Metabolic Cost).

During each trial that involved wearing the hip exoskeleton, angular position and filtered velocity of the left and right actuator encoders, 3-axis acceleration and 3-axis angular velocity data measured from inertial measurement units (IMUs) mounted on the left and right thigh struts as well as on the backplate, estimated hip moments from the TCN, commanded actuator torque, and measured actuator torque (computed by multiplying measured current from the motor drivers with the motor torque constant and actuator gear ratio) were recorded. All data from the exoskeleton were collected at 200 Hz. Additionally, when both motion capture and exoskeleton sensor data were recorded during the same trial, exoskeleton data were synced with the motion capture and GRF data by aligning the exoskeleton data to the rising edge of a 5V analog pulse generated by the Vicon Lock Sync Box at the trial start.

Phase 1: N = 9 Initial Dataset

To generate an initial dataset to train the hip moment estimator, we transformed the exoskeleton sensor data of a previously collected dataset (*35*) that used the Gait Enhancing and Motivating System (GEMS) (Samsung, South Korea), a lightweight, autonomous exoskeleton designed to assist the user's hip joints in the sagittal plane (*7*, *43*). The dataset consisted of nine able-body participants (Participants AB01-AB09 in table S1, 5 males, 4 females, average height of 1.75 ± 0.07 m, average body mass of 70.4 ± 8.7 kg, and average age of 22 ± 3 years) walking over level ground, inclines and declines of ±7.8°, ±9.2°, ±11.0°, and ±12.4°, and stairs of height 10.2 cm, 12.7 cm, 15.2 cm, and 17.8 cm while wearing the GEMS. The participants completed a minimum of 20 walking bouts over each condition while the exoskeleton provided hip assistance based on predefined gait phase-based spline trajectories. Gait phase was estimated onboard the device using a deep convolutional neural network validated in our previous work (*26*).

The GEMS data were collected at 200 Hz and consisted of angular positions and velocities of the actuator-mounted encoders and 3-axis acceleration and gyroscope data from left thigh-mounted, right-thigh mounted, and pelvis-mounted IMUs. The left and right thigh IMUs (MPU-9250, TDK InvenSense, CA, USA) were not installed on the GEMS by default but were manually added to improve the sensing capability of the device. Encoder velocities were computed post hoc using backward finite differencing and a 2nd order Butterworth filter with a lowpass cutoff frequency of 10 Hz to match the GEMS velocity signals to those of our custom hip exoskeleton. Additionally, the IMUs onboard the GEMS were placed on the same linkages as those mounted on our custom hip exoskeleton but were placed in different positions and orientations. To transform the data from the IMUs onboard the GEMS into the coordinate systems of those on our custom hip exoskeleton, we reoriented the data from each IMU using a 3-axis rotation and 3-axis translation optimized based on calibration data from a single participant (details provided in Transforming IMU Data from a Preexisting Exoskeleton Dataset for Implementation with a Novel Device).

In addition to the exoskeleton data, the dataset consisted of ground-truth lower-limb joint moments computed from motion capture (200 Hz) and GRF (1000 Hz) data. The joint moments were computed using the same methods described in the Biomechanical Modeling section. The resulting dataset included ~1.2 million labels consisting of left or right hip moments with corresponding exoskeleton data, which were added to the model training set used in this study.

Phase 2: N = 5 Overground Dataset

Since the dataset from Phase 1 was collected on a different exoskeleton, it was beneficial to collect training data from the specific device used in this study to minimize differences between the training set and testing set. Thus, we conducted a follow-up data collection with five able-body participants (Participants AB10-AB14 in table S1, 4 males, 1 female, average height of 1.73 ± 0.07 m, average body mass of 67.0 ± 3.3 kg, and average age of 25 ± 3 years). Participants walked overground under the same conditions as the dataset detailed in Phase 1, completing 10 "down-and-back" walking bouts per trial. To further expand the domain of the training set, the participants also completed 20 "down-and-back" walking bouts overground where one stand-to-walk and one walk-to-stand transition was recorded per bout. During the overground walking trials, the participants were instructed to walk with a starting leg pattern that collected an even number of left and right leg strides for each condition, for instance, "complete the first five bouts starting with your right foot and the second five bouts starting with your left foot." Finally, an additional trial of neutral standing for 30 seconds was included, in which participants were encouraged to slightly sway side-to-side and back-and-forth to further increase the richness of the data. During each trial, the exoskeleton provided assistance using the unified joint moment controller with a hip moment estimator trained on the Phase 1 dataset. Assistance was manually turned off by an experimenter during the standing trial and during the standing portion of the stand-to-walk and walk-to-stand transition trials during this phase of the experimental protocol since the model had not previously been trained on standing data. During each trial, time-synced data from the exoskeleton, motion capture system, and force plates were collected. Ground-truth hip moments were then computed as detailed in the Biomechanical Modeling section. The resulting Phase 2 dataset consisted of ~700,000 examples of labeled hip moments with corresponding exoskeleton data.

Phase 3: N = 10 User-Exoskeleton Training and User Outcomes Testing

Phase 3 of the experimental protocol was designed to test the effects of the unified joint moment controller on two user outcomes: the metabolic cost of walking and lower-limb positive mechanical joint work. Ten able-body participants (Participants AB15-AB24 in table S1, 7 males and 3 females) with an average height of 1.75 ± 0.07 m, body mass of 74.0 ± 12.6 kg, and age of 24 ± 3 years were enrolled in the protocol. The sample size was selected based on that of previous studies (*8*, *58*). The Phase 3 protocol was sectioned into two sessions. Session 1 trained the participants in walking with the exoskeleton, tested the effect of the exoskeleton assistance on each participant's lower-limb mechanical joint work, and provided an additional dataset for training and evaluating the hip moment estimator. Session 2 quantified the effect of the exoskeleton assistance on each participant's metabolic cost of walking. During this Phase 3 protocol, the unified joint moment controller was implemented using a hip moment estimator trained using the labeled data from Phase 1 and Phase 2.

During Session 1, each participant walked on a level treadmill (0.75, 0.8, 1.0, 1.2, 1.4, 1.5, 1.6, and 1.75 m/s), on an inclined/declined treadmill (-5 at 1.25 m/s, ±7.5° at 1.13 m/s, ±10° at 1.0 m/s, ±12.5° at 0.88 m/s, and ±15° at 0.75 m/s), and over a height-adjustable staircase (stair heights of ±10.2 cm, ±12.7 cm, ±15.2 cm, and ±17.8 cm) while wearing the hip exoskeleton (fig. S5), aside from Participants AB16-AB19 who did not complete the 0.75, 1.5, and 1.75 m/s level walking trials. During each condition, the exoskeleton provided hip flexion/extension assistance using the unified joint moment controller. Each treadmill condition lasted 30 seconds and each stair condition consisted of 12 passes up and down the 6-step staircase. Additionally, each participant completed treadmill trials of level walking and 5° ramp ascent both at 1.25 m/s while the exoskeleton provided assistance using the unified joint moment controller (Unified Control) and without wearing the exoskeleton (No Exo). Each condition lasted 6 minutes to ensure the participants' joint kinetics converged to a steady-state, with the last two minutes of each trial being used to analyze the exoskeleton effect on user lower-limb joint work (see Analyzing Exoskeleton Effects on User Mechanical Joint Work for more details). The Unified Control and No Exo conditions were completed in a randomized order as were the ambulatory conditions (between level walking and 5° ramp ascent); however, either both of the Unified Control conditions or both of the No Exo conditions were completed first to minimize don/doff time in the protocol. Time-synced data from the exoskeleton, motion capture system, and force plates were collected for all trials completed in Session 1 of the Phase 3 protocol. Thus, the Session 1 protocol allowed the participants to train with the exoskeleton, an important factor when studying exoskeleton effects on user metabolic cost (*50*), quantified the effects of the unified controller on lower-limb joint work, and yielded a third labeled dataset for hip moment estimation (~3 million examples of labeled hip moments).

Session 2 of the Phase 3 protocol was conducted to evaluate the effect of our exoskeleton controller on user metabolic cost. Volumetric flowrates of oxygen intake $(\dot{V}O_2)$ and carbon dioxide exhaust $(\dot{V}CO_2)$ from the breaths of each participant were measured and used to compute metabolic cost (details in Analyzing Exoskeleton Effects on User Metabolic Cost). The metabolic trials were conducted under the same ambulatory conditions used to analyze lower-limb joint work: walking at 1.25 m/s on a level treadmill and walking at 1.25 m/s on a 5° inclined treadmill. For each ambulatory condition, we tested four exoskeleton assistance conditions: wearing the exoskeleton which provided assistance based on our unified joint moment controller (Unified Control); wearing the exoskeleton which provided assistance based on previously optimized splines (Spline Control) (*20*, *21*); without wearing the exoskeleton (No Exo); and wearing the exoskeleton which commanded zero torque (Zero Torque). The assistance trajectories used for Spline Control were generated using piecewise cubic Hermite interpolating polynomials based on the average splines resulting from previous human-in-the-loop optimization studies for level ground walking and 5° ramp ascent at 1.25 m/s (*20*, *21*). The splines were parameterized by six nodes, describing the timing for extension onset, peak extension torque, extension offset, flexion onset, peak flexion torque, and flexion offset with respect to gait phase. Since the metabolic trials were conducted at steady-state, gait phase was computed at each point in time based on the average stride duration of the previous 20 strides. Each stride duration was computed based on measured heel strikes detected by force sensitive resistors mounted under the user's heel and integrated into our hip exoskeleton. The FSR data were also used to segment the users strides for analyzing the exoskeleton data from the metabolic trials post hoc. The assistance magnitude of Spline Control was modified for each participant to match the peak torque provided by the unified controller (Fig. 2c and 2f).

Before completing the metabolic trials, participants habituated to the exoskeleton by walking with assistance at 25, 50, and 75% of the final assistance level used for the metabolic trials, each lasting two minutes. Participants then walked with 100% of the final assistance level for 5 minutes. These habituation trials were completed with the powered assistance condition (either Unified Control or Spline Control) that was second in the experimental order. After this first habituation phase, participants then completed a final 5-minute habituation trial with the alternative assistance condition at 100% of the final assistance level. Thus, the participants walked for approximately 40 minutes with exoskeleton assistance from the Session 1 and 2 protocols before completing the metabolic trials. As, Poggensee *et al.* (*50*) found when using an ankle exoskeleton, user training time and curriculum can have substantial effects on metabolic cost. Though the participants were given substantial amounts of time to adapt to the exoskeleton assistance in this study, we did not measure metabolic cost during the training and habituation periods to reduce experimental complexity. Therefore, user metabolic adaptation rates to hip exoskeleton assistance and how it is affected by joint moment-based controllers opposed to spline-based controllers remains an open research topic.

Due to scheduling constraints Participants AB17 and AB18 completed Session 2 prior to Session 1, meaning they only experienced 16 minutes of exoskeleton walking before completing the metabolic trials; however, neither participant responded out of the norm (metabolic cost results for each participant are provided in the Supplementary Data).

The metabolic protocol used a within-participant counter-balanced design (ABCD-DCBA) with each trial lasting 6 minutes to allow the participants' metabolic cost to reach and maintain a steady-state (*8*, *60*). Participants were blinded to the exoskeleton assistance conditions, and the order of conditions were pseudo-randomized with the No Exo condition either in the A or D position to minimize don/doff time. The order of ambulation modes was also randomized, either completing all level ground walking trials or all ramp ascent trials first.

Phase 4: N = 10 Validation of the Hip Moment Estimator In-The-Loop

After conducting the Phase 3 experimental protocol, we noticed that TCN model performance substantially improved when adding the Phase 3 data into the model training set. This meant that the online performance of the TCN measured during the Phase 3 protocol was likely confounded by limitations in dataset size, preventing a thorough validation of the hip moment estimator for online performance. Thus, we conducted Phase 4 of the experimental protocol to quantify the accuracy of the hip moment estimator when implemented in the control loop and when trained on a substantially larger training set (the labeled datasets collected in Phases 1, 2 and 3). It is important to note that metabolic cost and joint work analyses were not reevaluated with the updated model since an improvement in model performance was very unlikely to change the results of the tested hypotheses. Notably, the unified controller significantly reduced user metabolic cost relative to No Exo and Zero Torque and significantly reduced lower-limb positive mechanical work relative to No Exo using the less accurate, 14-participant model in Phase 3.

Ten able-body participants (Participants AB25-AB34 in table S1, 8 males and 2 females) with an average height of 1.71 ± 0.08 m, body mass of 73.3 ± 12.5 kg, and age of 25 ± 5 years were enrolled in this phase of the protocol. Each participant walked with the hip exoskeleton on a level treadmill at 0.75, 0.8, 1.0, 1.2, 1.25, 1.4, 1.5, 1.6, and 1.75 m/s, on an inclined/declined treadmill at ±5° at 1.25 m/s, ±7.5° at 1.13 m/s, ±10° at 1.0 m/s, ±12.5° at 0.88 m/s, and ±15° at 0.75 m/s, and over a height-adjustable staircase with stair heights of ±10.2 cm, ±12.7 cm, ±15.2 cm, and ±17.8 cm (fig. S5). To analyze the online performance of the TCN when tested on novel conditions that were not included in the training set, each participant also walked on a level treadmill at 0.6, 1.1, 1.3 and 1.9 m/s and on an inclined/declined treadmill of ±6.3° at 1.19 m/s and ±13.8° at 0.82 m/s. Each participant also repeated the stair ascent/descent trials with stair heights of ±12.7 cm and ±17.8 cm while using a hip moment estimator with these two conditions iteratively withheld from the training set, respectively. Each treadmill trial lasted 30 seconds, and the stair ascent/descent trials consisted of completing six passes up and down the 6-step staircase. Finally, five of the 10 participants completed an additional five stand-to-walk and walk-to-stand transitions by ramping the treadmill from neutral standing to 1.25 m/s at 0.3 m/s^2 and from 1.25 m/s to neutral standing at -0.3 m/s^2, respectively. Stand-to-walk trials were segmented from the first toe-off in the trial to the first toe-off of the same leg after 4.17 seconds, which was the time it took the treadmill to accelerate to 1.25 m/s. Similarly, the walk-to-stand trials were segmented based on the last heel strike in the trial and the first heel strike of the same leg that occurred at least 4.17 seconds earlier. These participants also completed a 30 second trial of neutral standing. During each trial, the exoskeleton provided assistance using the unified joint moment controller based on the instantaneous hip moment estimates from the TCN. Throughout this protocol, exoskeleton, motion capture, and GRF data were collected using the methods described above and ground-truth joint moments were computed as described in the Biomechanical Modeling section. Due to a malfunction when syncing the exoskeleton and motion capture system, data from Participant AB25 was omitted for the 12.7 cm stair height trial that consisted of withholding this condition from the training set. Additionally, due to exoskeleton malfunctions, the 1.9 m/s level ground walking trial was not collected for Participant AB25. Finally, the -15° ramp descent condition for Participant AB28 was omitted due to a failure in the treadmill lock, which resulted in substantial noise in the force plate data. The resulting dataset consisted of ~3.5 million examples of labeled hip moments, which were used as the test set for evaluating the TCN.

Exoskeleton Hardware and Electronics

In this study, we used a custom-designed, autonomous hip exoskeleton (fig. S1) to validate the unified joint moment controller (key design specifications provided in table S2). The device provided sagittal plane hip assistance to the user via a high power-density, fully integrated actuator (AK80-9, T-Motor, China) with a 9 to 1 gear ratio and peak assistance torque of 18 Nm (~30% of peak biological hip moments during walking (*40*)). The low gear ratio of the actuators minimized their reflected inertia, making the system easy to backdrive in the case that the user needed to manually drive the device, for instance if the exoskeleton unexpectedly turned off (*61*). Additionally, the actuators were rated for a maximum speed of 35.6 rad/s at peak torque, far exceeding peak angular velocities of the hip joint in both walking and running (*40*, *62*).

Exoskeleton assistance was transmitted to the user's torso through a carbon fiber backplate and to the user's thighs via carbon fiber thigh struts. The backplate was coupled to a pelvis interface which was used to mount the actuators coaxially with the user's hip joints. The backplate was slotted, allowing mediolateral adjustability of the pelvis interface. Two tensioning belts secured the backplate and pelvis interface to the user's pelvis and torso. To ensure the device did not limit the user's range of motion, the actuator mounts were designed to accommodate up to 127° of hip flexion and 37° of hip extension. A passive hinge joint was also integrated into the thigh strut to accommodate hip ab/adduction. The participants did not reach the maximum range of motion of the device during any trials within this study.

A running backpack (Running Pack, Fitly, USA) housed a microprocessor (myRio, National Instruments, TX, USA), deemed the main processor, which executed the exoskeleton control loop at 200 Hz. The main processor communicated with the actuators via CAN bus. Since the main processor did not natively support CAN communication, we developed a custom SPI-to-CAN circuit using a SPI-CAN converter (MCP2515, Microchip Technology, AZ, USA) and CAN transceiver (TJA1050, NXP USA, TX, USA) to convert data packets between the main processor and the actuators. The circuit was soldered to a custom-designed PCB for robust integration on the device. In addition to the actuators, the main processor also obtained acceleration and angular velocity measurements via SPI from three IMUs (MPU-9250, TDK InvenSense, CA, USA) mounted on the backplate and left and right thigh struts. Finally, a second microprocessor (Jetson Nano, NVIDIA, CA, USA), deemed the co-processor, was also mounted to the electronics backpack and was used for rapid inference of the hip moment estimator (details provided in the Exoskeleton Software section).

The actuators and main processor were powered by a 24 V, 3600 mAh lithium polymer (LiPo) battery (Venom Fly, Venom Power, USA). The battery voltage delivered to the main processor was stepped down to 11 V using an adjustable step-down voltage regulator (LM2596, Texas Instruments, TX, USA). The co-processor was powered by a separate 5 V battery to reduce the total load on the LiPo battery; however, we did not need to replace either battery during the experiments, suggesting the 5 V battery may be removed with future improvements to the device. All components used to run the exoskeleton were mounted onboard the device to ensure the exoskeleton was completely autonomous.

Exoskeleton Software

The software used to control the exoskeleton was distributed across the main processor (the myRio) and the co-processor (the Jetson Nano) to efficiently integrate the neural network-based hip moment estimator into the control loop, as shown in fig. S2. Software on the main processor was developed using LabVIEW v2019, while the co-processor software was developed in Python v3.6.9. With each loop of the controller, the main processor received the angular position of the actuator output shaft from the actuator-mounted encoders and 6-axis acceleration and angular velocity data from the IMUs. Actuator velocity was computed onboard the main processor using backward finite differencing and was lowpass filtered using a 2nd order Butterworth filter with a 10 Hz cutoff frequency. Encoder and IMU data were subsequently streamed to the co-processor over an ethernet connection using TCP/IP.

To improve system robustness and reduce the latency of generating hip moment estimates, the co-processor software was segmented into two independent processes; one process was dedicated to input/output operations and managing data sequences (the I/O process), while the other was dedicated to model inference (the inference process). With each new message, the incoming sensor data were added to a first-in-first-out buffer on the co-processor via the I/O process. Input sequences for the left and right leg were then queried from the buffer and transformed (see transformation details in the following sections) such that two input sequences, one for each anatomical side, were generated from the latest exoskeleton sensor data. The input sequences were then concatenated into a 3D tensor with a batch size of two and passed to the inference process via a Python multiprocessing queue. The input tensor was copied to the GPU of the co-processor and passed through the model to estimate the left and right hip flexion/extension moments. To minimize inference latency of the estimator in real-time, the pretrained neural network was loaded onto the GPU of the co-processor after being converted into a TensorRT runtime engine using TensorRT v8.0.1.6. The resulting hip moment estimates were then copied to the co-processor CPU within the inference process, passed to the I/O process via a second multiprocessing queue, and returned to the main processor to be used for exoskeleton control via the same wired network. Using the parallelized framework, the I/O and queue operations were implemented with non-blocking logic based on predetermined timeouts, maintaining the exoskeleton loop rate even if the operations on the co-processor were too slow. In other words, if new data was not available fast enough, the process would continue with the data from the previous loop. The average latency between sending exoskeleton sensor data to the co-processor and receiving the corresponding hip moment estimates was 5 ms, meaning the estimated hip moments were only one timestep delayed on average.

The instantaneous hip moment estimates were then passed to the mid-level control layer implemented within the main processor control loop, which transformed the estimated hip moments into desired exoskeleton assistance torque (details provided in the Materials and Methods). Finally, the desired torque was passed to the motor drivers (the low-level layer), which converted the desired torque to commanded motor current using closed-loop current-feedback control.

To communicate with the main processor, a graphical interface was developed to display exoskeleton state information streamed from the main processor and to allow the experimenters to start the main processor software and toggle assistance. The co-processor communicated with the laptop using Secure Shell Protocol (SSH), allowing the experimenters to start the co-processor software from a remote terminal. The main processor, co-processor, and laptop were connected using a wireless network hosted by a nearby Wi-Fi router, removing the need for any wired connections with the laptop.

Temporal Convolutional Network Hyperparameters and Training

The TCN used to estimate hip moments in this study used the hyperparameters optimized in our previous work (*34*). Specifically, the model was comprised of five residual blocks each made up of two 1D causal convolutional layers that were each followed by a weight normalization layer and a ReLU activation (fig. S3). Dilated convolution exponentially increases the fixed input sequence length of the model with a linear increase in network depth (fig. S3c), minimizing the total number of model parameters compared to conventional convolutional neural networks. Specifically, the dilation factor (d) of the 1D convolutional filters in the i_{th} residual block of the network, indexed starting at 0, was computed as

$$d_i = 2^i, \tag{S1}$$

such that the 1D convolution operation $F(\cdot)$ returned the j_{th} output of the input sequence x as

$$F_j(x) = \sum_{n=0}^{k-1} f_n x_{j-dn}, \tag{S2}$$

where k and f_n are the kernel size and the n_{th} weight of the learned convolutional filter, respectively. Thus, the input sequence length (h) of the TCN was computed as

$$h = 1 + \sum_{i=0}^{l-1} 2(k-1)d_i, \tag{S3}$$

where l is the number of residual blocks of the network.

Each convolutional layer used 50 filters and a kernel size of four. The output of the previous residual block was summed elementwise with the output of the following residual block, which stabilized the network during training (*63*). An additional convolutional layer with kernel size of one was used to reshape the data passed via the residual connection in the case that the number of filter maps did not match from one residual block output to the next, which occurred at the first residual connection. A single linear layer was also implemented after the last residual connection of the TCN to compress the 50-channel output into a scalar value (the estimated hip moment).

The TCN was trained via supervised learning in Pytorch v1.8.0 using Python v3.6.9 on the Georgia Tech Pace Computing Cluster. Left-leg and right-leg training data sequences were randomly shuffled to train the model to estimate the hip moment from either anatomical side. Specifically, data from the left leg were mirrored such that the left leg input data used the same anatomical coordinate system as the right leg. Further, when inputting data for left-side hip moment estimation, the pelvis IMU data were also mirrored to fit the left-side anatomical coordinate system. This allowed a single network to be trained in a side-independent manner, meaning input sequence data pertaining to the left and right hip joint did not need to be distinguished.

Model training used the Adam optimizer with an MSE loss function and an initial learning rate of 0.0005 for a minimum of 100 epochs and maximum of 500 epochs based on our previous optimization (*34*). To minimize model training time, each mini-batch consisted of 4 sequences with a length of 218 (containing valid labels for the last 32 instances in each sequence). Early stopping based on data from a hold-out participant was used to end network training if the validation MSE did not decrease for 50 sequential epochs. In this case, the learned parameters that performed best on the validation data were reloaded and saved as the final model used for further analyses. The ground-truth label was the hip flexion/extension moments computed from our biomechanical model (details in the Biomechanical Modeling section) and scaled by each participant's body mass.

Biomechanical Modeling

All biomechanical analyses were conducted using the opensource musculoskeletal modeling software, OpenSim (*54*, *55*) with the gait2392 lower-limb model. Since the hip exoskeleton occluded markers placed on the pelvis, the segment dimensions, masses, and inertias of the default model were first scaled to each participant's anthropometry using motion capture data from a No Exo neutral standing trial. This scaled model was then used for computing joint kinematics and kinetics for trials without the exoskeleton. Since the exoskeleton occluded the motion capture markers around the pelvis, a copy of the scaled model was also rescaled using marker data from a second neutral standing trial consisting of the participant wearing the exoskeleton with a modified marker set. During the rescaling process, the OpenSim model segment masses and inertias were not modified (only marker locations relative to the segments were modified), which preserved the anthropometric features of each participant. The added mass of the exoskeleton was then added as a point mass to the pelvis of the rescaled model to minimize residual errors between the OpenSim model dynamics and the measured GRFs. The rescaled model was used to compute joint kinematics and kinetics for all trials that included wearing the exoskeleton.

Joint kinematics were computed for each trial using the OpenSim Inverse Kinematics tool based on the motion capture marker data. The resulting joint kinematics and corresponding GRFs for each trial were then used to compute joint moments using the OpenSim Inverse Dynamics tool. Unless otherwise stated, exoskeleton torque was not subtracted from the resulting hip flexion/extension moments, meaning these values represented the total torque at the joint (the torque from both the biological joint and the exoskeleton). Before computing joint kinematics and moments, marker data were lowpass filtered using a zero-lag, 5th order Butterworth filter with a 6 Hz cutoff frequency, and GRF data were lowpass filtered using a zero-lag, 5th order Butterworth filter with a 20 Hz cutoff frequency. Additionally, the resulting joint kinematics and moments were filtered using a zero-lag, 5th order Butterworth filter with a 6 Hz cutoff frequency.

Root mean square (RMS) root residuals were also averaged across all trials in the dataset. The average RMS residual forces were 39.76 ± 11.96 N, 56.93 ± 10.66 N, and 35.28 ± 7.74 N, and the average RMS residual moments were 20.97 ± 4.23 Nm, 9.43 ± 2.63 Nm, and 23.44 ± 6.37 Nm for the x-, y-, and z-axes relative to the pelvis. Based on OpenSim guidelines for residual magnitudes when modeling full-body walking, the RMS moments fell within the recommended range, but the RMS forces were larger than recommended. The large RMS residual forces were likely a result of unmodeled dynamics at or above the pelvis, such as arm swing. In general, unmodeled dynamics at or above the pelvis will affect residual forces and moments but will not affect lower-limb joint moments based on the methods used within the

OpenSim Inverse Dynamics tool. Further, the resulting hip moments from our biomechanical model were consistent with the results of previous studies (*40*, *64*). Therefore, the magnitudes of the residual moments and forces were deemed acceptable for the purposes of this study.

Analyzing Exoskeleton Effects on User Metabolic Cost

For each trial in which $\dot{V}O_2$ and $\dot{V}CO_2$ data were collected, instantaneous metabolic cost (P_{met}) scaled by body mass (m) was computed based on (*8*, *65*) as

$$P_{met} = \frac{\left(0.278 * \dot{V}O_2 + 0.075 * \dot{V}CO_2\right)}{m}. \tag{S4}$$

Steady-state metabolic cost was then computed as the average metabolic cost from the last three minutes of each six-minute trial (*8*, *60*). The metabolic cost of walking was computed by subtracting the participant's basal metabolic rate, computed as the steady-state metabolic cost during neutral standing, from the steady-state metabolic cost for each trial, a common method of evaluating the whole-body energetic cost for human movement (*3*). Since each assistance condition (Unified Control, Spline Control, No Exo, and Zero Torque) was run twice using the counter-balanced experimental design, the metabolic cost of walking for each participant was computed by averaging the results of the two trials for each assistance condition and ambulation mode.

Analyzing Exoskeleton Effects on User Mechanical Joint Work

Sagittal plane net joint power (P_{net}), the total power at the joint including both human and exoskeleton contributions, was computed at the hip, knee, and ankle using the angular velocity of the biological joint $\left(\dot{\theta}_{bio}\right)$ and the net joint moment (τ_{net}) as

$$P_{net} = \tau_{net} * \dot{\theta}_{bio}. \tag{S5}$$

Similarly, the exoskeleton power delivered to the biological joint (P_{exo}) was computed as

$$P_{exo} = \tau_{exo} * \dot{\theta}_{bio}, \tag{S6}$$

such that the human-exoskeleton interaction torque (τ_{exo}) was computed by subtracting the actuator torque lost to motor dynamics $\left(\tau_{dyn}\right)$ from the total actuator torque (τ_{act}) as

$$\tau_{exo} = \tau_{act} - \tau_{dyn}. \tag{S7}$$

Since the actuators used a low gear ratio (N) of 9 to 1, τ_{act} was computed from the measured motor current (i_{motor}) and motor torque constant (k_t), such that

$$\tau_{act} = N * k_t * i_{motor}. \tag{S8}$$

Additionally, τ_{dyn} was computed from the angular velocity $\left(\dot{\theta}_{act}\right)$ and acceleration $\left(\ddot{\theta}_{act}\right)$ of the actuator output encoder using a second-order model of the actuator dynamics, as

$$\tau_{dyn} = N^2 * \left(J * \ddot{\theta}_{act} + b * \dot{\theta}_{act}\right), \tag{S9}$$

where the motor inertia (J) and damping constant (b) were approximated from a similar electric motor (*66*) ($J = 1.2 \times 10^{-4}$ kg·m^2 and $b = 1.6 \times 10^{-4}$ Nm·s). Finally, the biological joint power of the user (P_{bio}) was computed as

$$P_{bio} = \tau_{bio} * \dot{\theta}_{bio}, \tag{S10}$$

where the biological joint moment (τ_{bio}) was computed based on τ_{net} and τ_{exo}, as

$$\tau_{bio} = \tau_{net} - \tau_{exo}. \tag{S11}$$

Average positive mechanical work per joint (W^+) was computed by averaging the positive mechanical work of each of the S strides collected per condition (W_i^+), such that

$$W^+ = \frac{1}{S}\sum_{i=1}^{S} W_i^+, \tag{S12}$$

where the positive mechanical work of the i^{th} stride with duration d was computed from the positive power over the stride (P_i^+), as

$$W_i^+ = \int_0^d P_i^+(t)\, dt, \tag{S13}$$

with the positive power at each time instance t computed as

$$P^+(t) = \begin{cases} P(t) & if\ P(t) > 0 \\ 0 & otherwise \end{cases}. \tag{S14}$$

Total positive lower-limb joint work for each stride was then computed by summing the average positive joint work at the hip, knee, and ankle. All results are presented as the inter-participant average of the resulting positive work values.

<u>Transforming IMU Data from a Preexisting Exoskeleton Dataset for Implementation with a Novel Device</u>

In this study, we trained a preliminary version of the hip moment estimator based on the nine-participant dataset collected for our previous work (*35*). This allowed the unified joint moment controller of our study to be deployed immediately, without having to collect additional training data using alternative control strategies. The preexisting dataset was collected using the Gait Enhancing and Motivating System (GEMS), an autonomous robotic hip exoskeleton designed to provide sagittal plane hip assistance (Samsung, South Korea). The dataset included ground-truth joint moments of the user during overground walking with time-synced exoskeleton sensor data measured from an integrated pelvis IMU, integrated actuator encoders, and two IMUs custom-mounted on the thigh struts. Thus, the dataset included the same sensor modalities as those of our custom hip exoskeleton but the positions and orientations of the IMUs were inconsistent between the two devices. Since the GEMS and our custom hip exoskeleton fit the user differently, we did not assume transformations between the IMUs of the GEMS and the IMUs of our custom hip exoskeleton could be measured directly. Further, the exact position and orientation of the GEMS pelvis IMU was unknown since it was mounted internally in the pelvis support structure. Instead, we optimized a rotation matrix (${}^{E1}\hat{R}^{*\,E2}$) and position vector (${}^{E1}\hat{p}^{*}_{E2E1}$) mapping the coordinate system of each GEMS (EXO2) IMU to that of our custom exoskeleton (EXO1).

One participant (Participant AB10 in table S1, male, height of 1.77 m, body mass of 66.1 kg, and age of 29 years) walked on a treadmill at 1.3 m/s in EXO1 and EXO2 with assistance turned off while exoskeleton sensor data was recorded. To further improve the similarity in joint kinematics between the two conditions, the participant was asked to match his step cadence to a 110 bpm metronome. Ten seconds of exoskeleton sensor data from EXO1 and EXO2 were then collected in sequence and synced in time with each other by aligning the peak hip extension joint angle measured by the right-side hip actuator encoders. The resulting accelerometer and gyroscope data of EXO1 and EXO2 (A_{E1}, A_{E2}, G_{E1}, G_{E2}, respectively) were filtered using a zero-lag 5$^{\text{th}}$ order lowpass filter with a 20 Hz cutoff frequency to remove high frequency noise that would disrupt the similarities between the EXO1 and EXO2 IMU signals. The accelerometer data (A) and gyroscope data (G) were structured with the form

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,N} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,N} \\ a_{3,1} & a_{3,2} & \cdots & a_{3,N} \end{bmatrix} \tag{S15}$$

and

$$G = \begin{bmatrix} g_{1,1} & g_{1,2} & \cdots & g_{1,N} \\ g_{2,1} & g_{2,2} & \cdots & g_{2,N} \\ g_{3,1} & g_{3,2} & \cdots & g_{3,N} \end{bmatrix}, \tag{S16}$$

in which the x-, y-, and z-axes of the sensors are represented in each row and N represents the total number of timesteps in the sequence. To align the EXO1 and EXO2 IMU data, we first defined the $\mathbb{R}^6$ vector Z, which was comprised of three Euler angles $(\hat{\theta}, \hat{\phi}, \hat{\psi})$ and the position vector $(\hat{p})$ as

$$Z = \begin{bmatrix} \hat{\theta} \\ \hat{\phi} \\ \hat{\psi} \\ \hat{p} \end{bmatrix}, \tag{S17}$$

where the estimated rotation matrix (${}^{E1}\hat{R}^{E2}$) was computed from the three Euler angles as

$$\hat{R}(\theta, \phi, \psi) = R_z(\hat{\psi}) * R_y(\hat{\phi}) * R_x(\hat{\theta}). \tag{S18}$$

Using the estimated rotation matrix, the EXO2 gyroscope data was transformed into the EXO1 coordinate system as

$$\hat{G}_{E1} = \hat{R} * G_{E2}. \tag{S19}$$

Similarly, the EXO2 accelerometer data was transformed into the EXO1 coordinate system based on rigid body kinematics as

$$\hat{A}_{E1} = A_{E2} + (\hat{\alpha}_{E1} \times \hat{p}) + (\hat{G}_{E1} \times (\hat{G}_{E1} \times \hat{p})), \tag{S20}$$

in which the angular acceleration of the EXO2 IMU in the EXO1 coordinate system ($\hat{\alpha}_{E1}$) was computed as

$$\hat{\alpha}_{E1} = \frac{d}{dt}\hat{G}_{E1}. \tag{S21}$$

From this formulation, the vector Z^* was optimized to minimize the mean-squared error (MSE) between the EXO1 and transformed EXO2 gyroscope data $(\tilde{G})$ and the MSE between the EXO1 and transformed EXO2 accelerometer data $(\tilde{A})$ as

$$Z^* = \begin{bmatrix} \hat{\theta}^* \\ \hat{\phi}^* \\ \hat{\psi}^* \\ \hat{p}^* \end{bmatrix} = \underset{\hat{\theta},\hat{\phi},\hat{\psi},\hat{p}}{\operatorname{argmin}} \left(\sum \tilde{G}(\hat{\theta}, \hat{\phi}, \hat{\psi}) + \sum \tilde{A}(\hat{\theta}, \hat{\phi}, \hat{\psi}, \hat{p}) \right), \tag{S22}$$

such that

$$\tilde{G} = \begin{bmatrix} \tilde{g}_1 \\ \tilde{g}_2 \\ \tilde{g}_3 \end{bmatrix}, \tag{S23}$$

where the MSE along each axis of the gyroscope $(\tilde{g}_i)$ was computed as

$$\tilde{g}_i = \frac{1}{N}\sum_{j=1}^{N}\left(g_{i,j} - \hat{g}_{i,j}\right)^2 \ \forall\ g_{i,j} \in G_{E1,i}, \hat{g}_{i,j} \in \hat{G}_{E1,i}, \tag{S24}$$

and such that

$$\tilde{A} = \begin{bmatrix} \tilde{a}_1 \\ \tilde{a}_2 \\ \tilde{a}_3 \end{bmatrix}, \tag{S25}$$

where the MSE along each perpendicular axis of the accelerometer $(\tilde{a}_i)$ was computed as

$$\tilde{a}_i = \frac{1}{N}\sum_{j=1}^{N}\left(a_{i,j} - \hat{a}_{i,j}\right)^2 \ \forall\ a_{i,j} \in A_{E1,i}, \hat{a}_{i,j} \in \hat{A}_{E1,i}. \tag{S26}$$

Using the time-aligned EXO1 and EXO2 IMU data, we optimized a unique rotation matrix and position vector for each IMU using the fmincon function in MATLAB and the constraints shown in table S3. After transforming the EXO2 gyroscope data using Eqn. S19, the optimized alignment RMSE compared to the EXO1 data was 0.35, 0.42, and 0.17 rad/s (compared to 2.13, 2.24, and 0.33 rad/s without transforming the data), for the left thigh, right thigh, and pelvis IMUs, respectively. Similarly, after applying the optimized transformation to the EXO2 accelerometer data using Eqns. S20 and S21, the alignment RMSE of the accelerometers was reduced to 2.26, 2.86, and 0.90 m/s^2 (compared to 12.50, 13.23, and 8.80 m/s^2 without transforming the data) for the left thigh, right thigh, and pelvis IMUs, respectively. Thus, by optimizing a rotation matrix and position vector to align each of the IMUs from the original dataset with those of our custom hip exoskeleton, the sensor data could be used for training the TCN as if it was collected from our device.

Effect of Delay Timing on User Metabolic Cost

To tune the delay of the mid-level controller, we conducted a small, three-participant pilot study (Participants AB10-AB12 in table S1, 3 males, average height of 1.76 ± 0.08 m, average body mass of 67.1 ± 3.7 kg, average age of 24 ± 4 years) in which user metabolic cost of walking was measured under five delay magnitudes when using the unified joint moment controller. The evaluated delays were 75, 100, 125, 150, and 175 ms including the delay from the lowpass filter. The order of delay magnitudes was randomized and blinded to each participant. Each delay magnitude was evaluated during level ground walking at 1.4 m/s and 5° ramp ascent at 1.0 m/s using a within-participant counter-balanced design (ABCDE-EDCBA). Each trial lasted four minutes while the volumetric flowrate of oxygen intake $(\dot{V}O_2)$ and of carbon dioxide exhaust $(\dot{V}CO_2)$ of the participant's breath were measured using a metabolic measurement system (TrueOne 2400, ParvoMedics, UT, USA). Each participant's basal metabolic rate was also measured during neutral standing without wearing the exoskeleton. Hip moment estimates used in the unified joint moment controller were estimated using a TCN trained on the data from Phase 1 of the experimental protocol.

Prior to the metabolic tests, participants underwent a habituation protocol to acclimate to the exoskeleton assistance. The participants first walked for 1 minute each with a scale factor of 7.5% and 15% of the total estimated hip moment and a delay magnitude of 125 ms (the median delay magnitude). The participants then walked with full exoskeleton assistance (20% of the estimated hip moment) with a delay magnitude of 125, 100, 75, 150, and 175 ms, sequentially for two minutes each. The habituation trials were completed under the same ambulatory condition (level ground or ramp ascent) as the following metabolic trials.

Instantaneous metabolic cost was computed as detailed in Analyzing Exoskeleton Effects on User Metabolic Cost. The resulting data was fit using a first-order model to estimate steady-state metabolic cost from the four-minute trial (*8*, *60*). The steady-state metabolic cost of walking was then computed by subtracting each participant's basal metabolic rate from the estimated steady-state metabolic cost.

In both the level ground walking and ramp ascent trials, the metabolic cost of walking showed little sensitivity to the delay magnitudes tested in this pilot study (fig. S4b). During level ground walking, the average metabolic cost only varied by 0.07 W/kg across conditions (2.6% of lowest value). Similarly, during ramp ascent the average metabolic cost varied by 0.11 W/kg across conditions (2.9% of lowest value). Though we did not find substantial differences in metabolic cost across delay magnitudes, it is likely that further decreasing the delay would further increase metabolic cost, as the largest metabolic cost for both level ground and incline walking was measured during the lowest delay magnitude of 75 ms. Delays below 75 ms were noted as less comfortable during initial pilot experiments by both novice and expert exoskeleton users, leading us to omit them from this pilot study.

Testing the Hip Moment Estimator on Novel Conditions

To predict the real-world performance of machine learning-based high-level state estimators, it is critical to evaluate these systems under novel conditions, such as walking speeds, inclines/declines, and stair heights that are not included in the training set. Otherwise, researchers may not detect overfitting, falsely assuming their models generalize across conditions. In our previous work, we found that withholding each ground slope and stair height from the training set had little effect on the RMSE and R^2 of our hip moment estimator outside of the steepest incline of 18° (*34*). Though these results were promising, our analysis relied on simulated IMUs and was conducted offline. Given these limitations, the TCN could fail to generalize to novel conditions when using real exoskeleton sensor data, which contains substantial amounts of noise from measurement uncertainty and the physical decoupling between the human and exoskeleton. Further, the effect on model generalization of implementing the hip moment estimator in the control loop remains unknown. It is possible that inaccuracies in the hip moment estimator during novel conditions could propagate as the resulting exoskeleton assistance based on incorrect estimates could drive the user into a gait pattern far outside of the training set distribution (*49*).

In this study, we tested the TCN when implemented online (deployed onboard the exoskeleton and used for exoskeleton control) during several ambulatory conditions that were included in the training set, denoted "Within Train Set, Tested Online;" however to evaluate the generalizability of our controller framework, we also tested the unified controller under completely novel conditions of level ground walking (0.6 m/s, 1.1 m/s, 1.3 m/s and 1.9 m/s) and ramp ascent/descent (±6.3° at 1.19 m/s and ±13.8° at 0.82 m/s) and during stair height conditions withheld from the training set (±12.7 cm and ±17.8 cm). The results from these level ground walking, ramp ascent/descent, and stair ascent/descent trials were denoted "Not Within Train Set, Tested Online." The level ground walking speeds were selected to test the TCN accuracy with never-before-seen walking speeds both within and outside of the training set distribution (training set ranged from 0.75 to 1.75 m/s). The novel incline and decline slope angles all fell within the training set distribution (training set ranged from -15° to +15°) as 15° was the maximum slope of the treadmill. To test online TCN generalization during stair ascent/descent, we could not select completely novel stair heights since the height-adjustable staircase only included four preset stair heights (spanning the full range of ADA compliant stair heights). Instead, two additional TCN models were trained from random initialization a priori: one while withholding the ±12.7 cm data from the training set and the other while withholding the ±17.8 cm data. These models were then deployed during separate trials of the experiment and tested on the respective stair heights withheld from each model.

To quantify TCN generalizability during the remaining conditions of the Phase 4 experimental protocol (nine level ground walking speeds, five inclines and declines, and two stair heights), we quantified the TCN performance offline using a leave-one-out approach. These results were denoted "Not Within Train Set, Tested Offline." Specifically, we conducted a leave-one-out analysis in which we iteratively withheld each ambulatory condition from the training set while training the TCN from random initialization. For each model, the labeled data from

Phases 1, 2, and 3 of the experimental protocol were used, which was the same labeled dataset used to train the models deployed online. Each model was trained using the same validation set for early stopping (the data from Participant AB05 of the Phase 1 dataset) and was tested on each respective hold-out condition from the Phase 4 dataset. To ensure the ambulatory conditions were completely withheld from the training set, the Phase 1 level ground walking data (overground at a self-selected walking speed) were removed when training the level ground walking hold-out models. Similarly, the ±7.8° Phase 1 ground slope data were removed for the 7.5° hold-out model, the ±9.2° and ±11° Phase 1 ground slope data were removed for the 10° hold-out model, and the ±12.4° Phase 1 ground slope data were removed for the 12.5° hold-out model.

Additionally, we conducted an offline leave-one-in analysis to test the effect of including the never-before-seen ambulatory conditions evaluated online in the Phase 4 protocol in the model training set. The results from these conditions were denoted as "Within Train Set, Tested Offline." Since these conditions did not exist in the Phase 1, 2, and 3 datasets, we retrained the TCN with the labeled data from Phases 1, 2, and 3 as well as with the data for each respective hold-in condition from the Phase 4 dataset. Since the models were subsequently tested on the Phase 4 dataset, we iteratively retrained the TCN using a leave-one-participant-out, hold-one-condition-in approach in which 10 models were trained for each hold-in condition, with one model per withheld participant. The 10 models were subsequently tested on the data corresponding to the hold-in ambulatory condition of their respective hold-out participant, maintaining complete participant-independence in the analysis. During training, the same validation set was again used for early stopping (the data from Participant AB05 of the Phase 1 dataset) to prevent overfitting.

Using this approach, we obtained an average RMSE and R^2 of the TCN for each ambulatory condition when both including and excluding it from the training set (fig. S7). To quantify any loss in model accuracy due to testing on novel conditions, we used a two-way ANOVA to test for significant main and interaction effects among the ambulatory conditions and between the hold-in and hold-out conditions for each ambulation mode. Upon finding a significant effect, pairwise comparisons within each ambulation mode were conducted using a post hoc multiple comparisons test with a Bonferroni correction. Of all possible pairwise comparisons, pairwise differences were only tested between the hold-in and hold-out results within each ambulation condition (further details about statistical analyses are provided in the Materials and Methods).

When testing the relevant pairwise comparisons, we found a significant difference in RMSE of the hold-out models compared to the hold-in results for the 5° ramp ascent condition (increase of 0.031 ± 0.022 Nm/kg [22.2 ± 16.0%]; P = 0.0245), for the +17.8 cm stair ascent condition (increase of 0.019 ± 0.024 Nm/kg [14.1 ± 17.9%]; P = 0.0265), and for the -17.8 cm stair descent condition (decrease of 0.017 ± 0.020 Nm/kg [11.1 ± 12.9%]; P = 0.0388) (fig. S7). All other comparisons of RMSE and R^2 between the hold-in and hold-out conditions were not statistically significant. The p-values of all comparisons conducted for this analysis are provided in the Supplementary Data.

Overall, the model generalized well to the conditions within the distribution of the training set (interpolation), which was consistent with our previous offline findings (*34*). Within the ramp ascent and stair ascent ambulation modes we found significant increases in RMSE during the conditions that had the largest range of peak-to-peak hip moments (5° ramp ascent at 1.25 m/s and +17.8 cm stairs; see Fig. 5 for representative strides). Thus, the TCN started to increase in estimation error at the bounds of the training set distribution. Interestingly, we did not find any significant differences in R^2 between the hold-in and hold-out conditions, suggesting the TCN correctly maintained the "shape" of the hip moment curves even when tested under these extreme conditions. Thus, even when encountering high intensity conditions outside of the training set distribution, the unified joint moment controller maintained the correct assistance timing but may have begun to incorrectly scale the assistance magnitude.

Opposite to our expectations, we found that the TCN RMSE of the -17.8 cm stair descent condition was lower (better) when withheld from the training set compared to when it was included in the training set. This result suggests that adding condition-specific data actually worsened model RMSE when tested on this specific condition. It is likely that the low signal-to-noise ratio of the exoskeleton kinematic sensors along with the heterogeneity in participant-to-participant hip moments during steep stair descent did not provide useful training examples for the TCN in this condition. Regardless, we found that the magnitude of the -17.8 cm SD RMSE of the hold-in and hold-out models were still similar to those of the other conditions; however, further analyses should be conducted, investigating additional sensors or alternative loss functions that may improve the usefulness of this training data.

The results of this analysis also underscore that the model maintained performance when implemented online. In fact, there were no conditions in which the online model performed significantly worse than its offline counterpart. Though discrete classification methods can have much larger online errors due to error propagation from stride-to-stride (*49*), this result demonstrates that small errors in regression-based methods for high-level state estimation have little effect on the resulting human-exoskeleton dynamics. This is a key insight when considering exoskeleton control in the real-world, as large errors in applied exoskeleton assistance, such as those from a misclassification of ambulation mode, could cause the user to stumble or fall. Further, these results suggest that offline analyses of regression models for joint moment estimation reflect online results, improving the possible throughput for analyzing these types of estimators.

<u>Effect of Training Data on Hip Moment Estimation Accuracy</u>

As it is with any data-driven approach, the accuracy of neural network models is heavily dependent on the number of samples available in the training dataset. Thus, we conducted an offline analysis to quantify the effect of iteratively increasing the size of the training dataset on TCN accuracy. We iteratively added data from each phase of the study protocol (Phases 1 through 4) into the training set, substantially increasing the amount of labeled data (and total number of participants) with each iteration. Thus, the TCN was tested when using four different training sets of varying sizes: using data from Phase 1 only (P1; n = 9 training set), using data from Phases 1 and 2 (P1+2; n = 14 training set), using data from Phases 1, 2, and 3 (P1+2+3; n = 24 training set) and using data from Phases 1, 2, 3, and 4 (P1+2+3+4; n = 33 training set). The models for each condition were trained from random initialization while withholding the data from Participant AB05, which was used as the validation set for early stopping during model training (see Temporal Convolutional Network Hyperparameters and Training for more details on model training).

The resulting models were each tested offline on the level ground, ramp ascent, ramp descent, stair ascent, and stair descent data from the Phase 4 dataset. Since Phase 4 data was also included in the training set for the P1+2+3+4 condition, we iteratively retrained the model from random initialization using a 10-fold leave-one-participant-out approach, in which the data of the held-out Phase 4 participant was used as the test set. This approach ensured that the model would not be trained and tested on data from the same participant while providing insight into the effects of using the Phase 4 dataset for model training.

Significant main effects in RMSE and R^2 among the training set conditions were tested for using a one-way ANOVA. Additionally, a post hoc multiple comparisons test with a Bonferroni correction was used to test for significant pairwise differences among the conditions. Finally, an exponential decay function $(y = C_1 e^{-C_2 x} + C_3)$ parameterized by the $\mathbb{R}^3$ vector (C) was fit to the RMSE and R^2 data using the nlinfit function in MATLAB to evaluate the expected half-life and asymptotic result on model performance with respect to dataset size.

As shown in fig. S8a, we found that the P1+2+3 condition significantly reduced test RMSE compared to the P1 condition (change of 0.055 ± 0.017 Nm/kg [27.4 ± 8.5%]; $P = 1 \times 10^{-8}$) and the P1+2 condition (change of 0.040 ± 0.019 Nm/kg [21.6 ± 10.2%]; $P = 4 \times 10^{-6}$). The P1+2+3+4 condition also significantly reduced test RMSE compared to the P1 condition (change of 0.068 ± 0.024 Nm/kg [34.0 ± 12.2%]; $P = 1 \times 10^{-10}$) and P1+2 condition (change of 0.053 ± 0.028 Nm/kg [28.8 ± 14.9%]; $P = 2 \times 10^{-8}$); however, we did not find a significant difference in RMSE between the P1 and P1+2 conditions or between the P1+2+3 and P1+2+3+4 conditions. The half-life of the RMSE exponential decay fit was 2.15×10^6 labels with a corresponding RMSE of 0.123 Nm/kg at the asymptote.

Further we found similar results in R^2 (fig. S8b), with the P1+2+3 condition significantly improving R^2 by 0.131 ± 0.039 (18.5 ± 5.6%) compared to the P1 condition ($P = 9 \times 10^{-10}$) and by 0.087 ± 0.041 (11.6 ± 5.4%) compared to the P1+2 condition ($P = 2 \times 10^{-6}$). The P1+2+3+4 condition also significantly improved R^2 compared to the P1 condition by 0.149 ± 0.056 (21.0 ± 7.9%; $P = 5 \times 10^{-11}$) and compared to the P1+2 condition by 0.105 ± 0.056 (14.0 ± 7.4%; $P = 8 \times 10^{-8}$). In this case, we also found that the P1+2 condition significantly improved R^2 compared to the P1 condition (change of 0.044 ± 0.022 [6.2 ± 3.2%]; $P = 0.0149$); however, no significant difference in R^2 was found between the P1+2+3 and P+1+2+3+4 conditions. The half-life of the R^2 exponential decay fit was 1.48×10^6 labels with a corresponding R^2 of 0.864 at the asymptote.

Thus, adding in the Phase 3 data to the training set significantly improved model performance compared to only using the Phase 1 and 2 data; however, we found that also including the Phase 4 data into the training set had diminishing returns, even though the data from Phase 4 consisted of an extra 3.1 million labeled training examples (an increase in training set size of 62.3%). Thus, the online TCN validation conducted in Phase 4 of the study protocol, which used a TCN trained on the P1+2+3 dataset, provides a rigorous analysis of the TCN when considering the training set size.

Additionally, based on the fit trendlines, we found that the expected model performance given an infinite number of training examples would only improve by 0.009 Nm/kg in RMSE and 0.007 in R^2, compared to training on all of the available data in our dataset (the P1+2+3+4 condition). In general, further increasing the labeled dataset size could further increase model performance; however, the diminishing returns in model performance from this analysis suggest that researchers can use this publicly available dataset to evaluate new approaches for estimating human biomechanical signals from wearable sensors without major concerns about the size of the dataset.

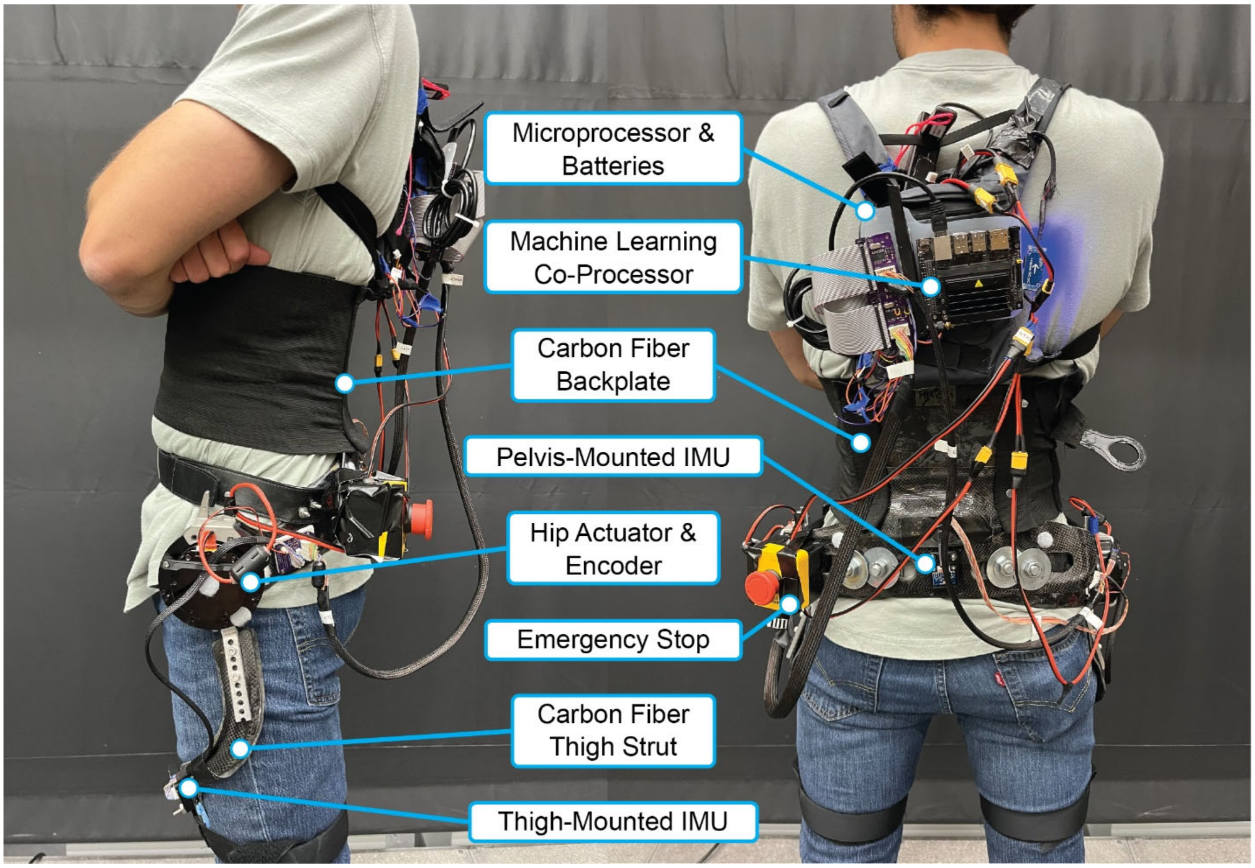

Fig. S1.

Autonomous robotic hip exoskeleton. The autonomous robotic hip exoskeleton used in this study was developed in-house. Assistance was provided to the user's hip joints in the sagittal plane by actuators mounted in parallel with the joint. The loads resulting from the actuators were transmitted to the user through custom interfaces made primarily of carbon fiber. The main processor for running the exoskeleton control was mounted on a running backpack with the machine learning co-processor, batteries, and custom PCBs, making the exoskeleton completely autonomous. Inertial measurement units (IMUs) were mounted on the thigh struts and backplate to provide sensor data in addition to the actuator encoders.

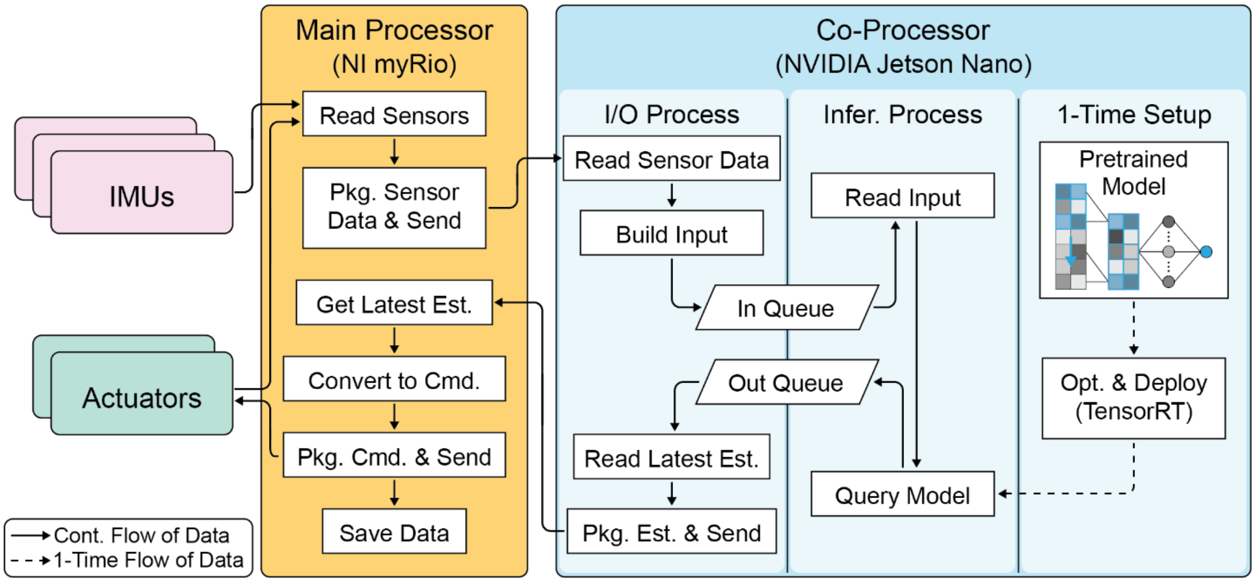

Fig. S2.

Flow diagram of exoskeleton software. Data from the inertial measurement units (IMUs) and actuator encoders were read by the main processor and immediately sent to the machine learning co-processor. The co-processor ran two independent processes. The first process handled I/O with the main processor and shaped the data for model inference. The second process was dedicated solely to model inference to minimize estimate latency. The resulting hip moment estimates from the model were then sent back to the main processor, which converted the resulting estimates into torque commands sent to the actuators. To further increase the inference speed, the neural network used to estimate the user's hip moments was converted to a TensorRT runtime engine before deployment. Though uncommon throughout the study, in the case that any of the processes on the co-processor ran too slowly, the I/O and queue operations within the main processor and co-processor were implemented with non-blocking logic to maintain the 200 Hz exoskeleton loop rate.

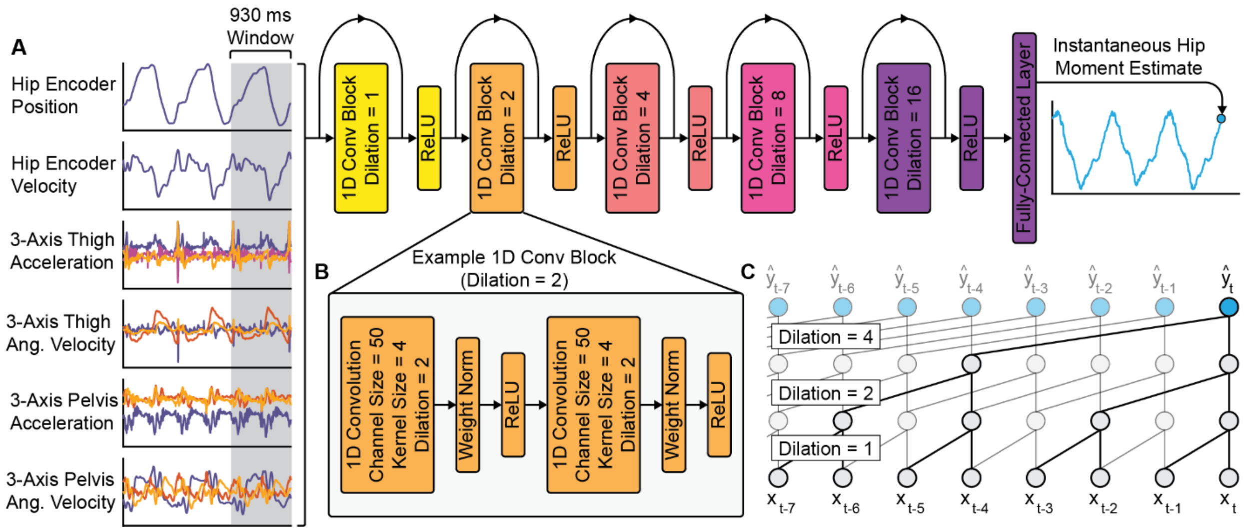

Fig. S3.

Temporal convolutional network for hip moment estimation. (**A**) The temporal convolutional network (TCN) used a 930 ms window of exoskeleton sensor data to estimate instantaneous hip flexion/extension moments. The dilation factor of the convolutional filters was exponentially increased with each subsequent residual block. (**B**) The residual blocks (1D Conv Blocks) were comprised of two 1D convolutional layers, each followed by a weight normalization layer and a ReLU activation function. Only the details of the second residual block of the network are shown; however, the remaining residual blocks were implemented using the same structure, only modifying the dilation factor as depicted. (**C**) A simplified graphic depicting the effect of convolutional kernel dilation is shown, in which x_t denotes the input data at time t, and $\hat{y}_t$ denotes the corresponding model output at the same timestep. In the simplified graphic, the dilation factor is increased with each subsequent layer; however, in the TCN used in this study the dilation factor was increased every two layers (with each new residual block).

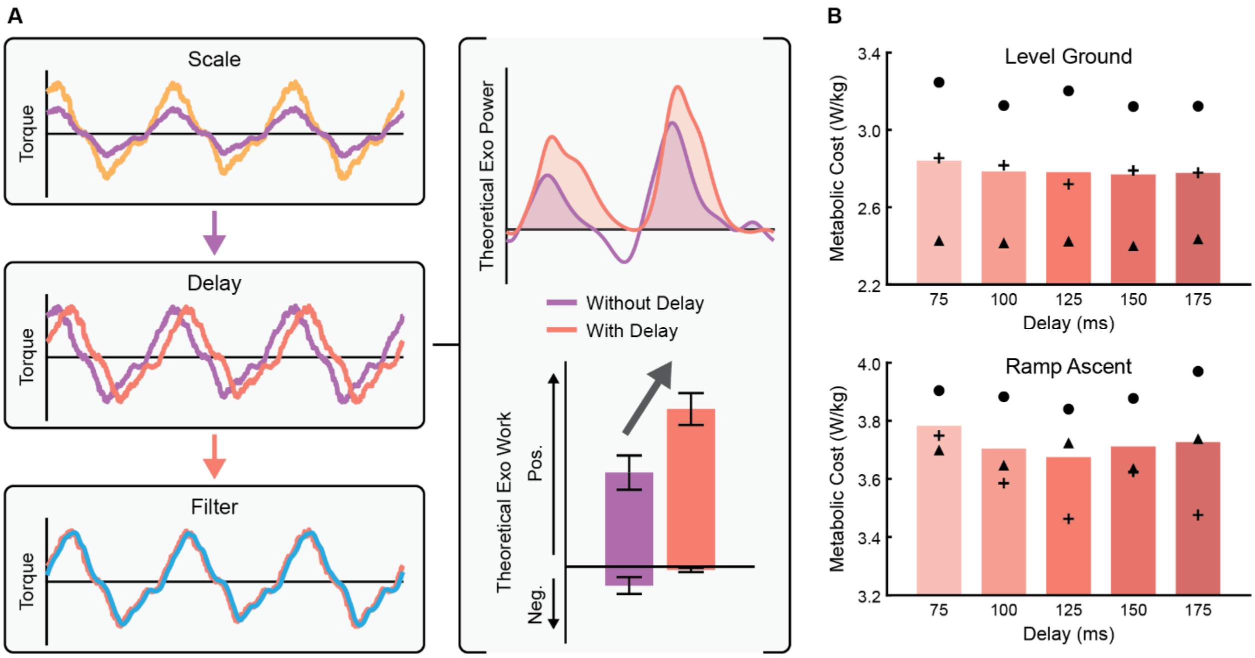

Fig. S4.

Mid-level layer of the unified joint moment controller. (**A**) The mid-level control layer computed desired exoskeleton assistance based on the instantaneous hip moment estimates from the high-level layer. As shown in the left column, the mid-level controller was implemented in three steps in which the hip moment estimates were scaled to a desired assistance percentage, delayed by a predefined magnitude, and lowpass filtered. As shown to the right, we expected that delaying the torque assistance relative to the estimated hip moments would substantially increase the positive mechanical work delivered from the exoskeleton to the user. The theoretical results were computed from the 1.25 m/s treadmill walking data published in (*40*). (**B**) The metabolic cost of walking of three participants during level ground walking and 5° ramp ascent was measured across several magnitudes of mid-level delay. Bars represent means, and markers represent the individual results of each participant. Statistical tests were not run given the small number of participants.

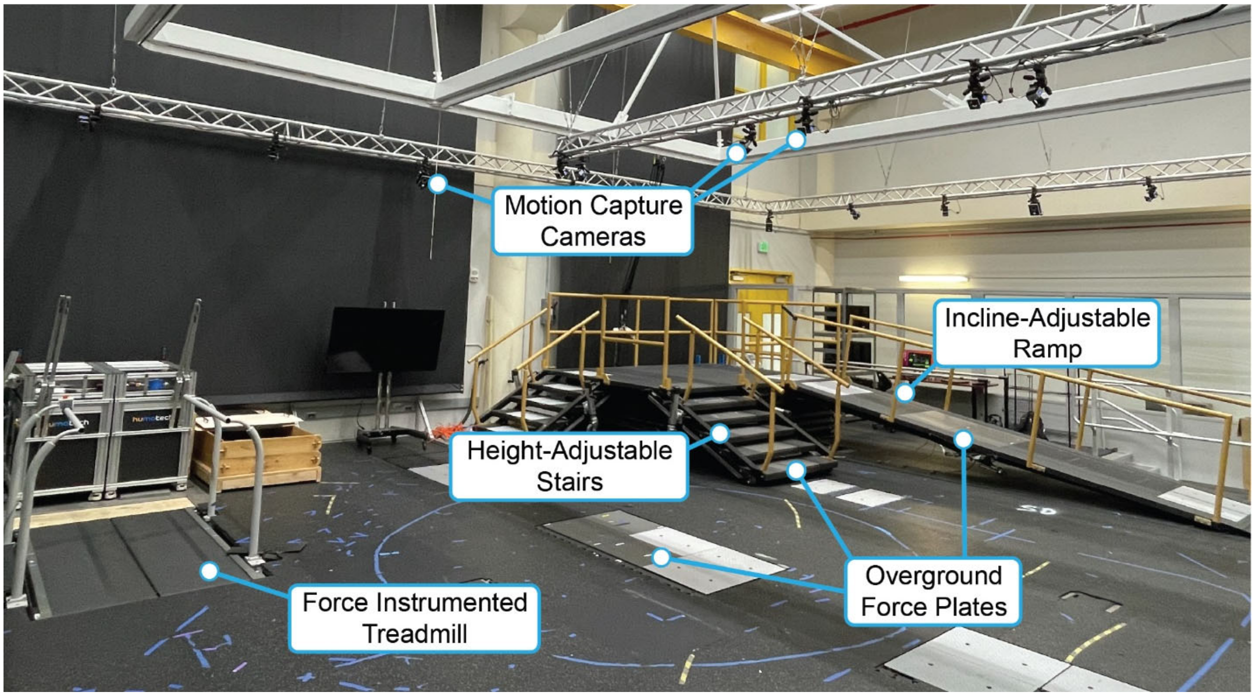

Fig. S5.

Experimental lab space. Ground-truth lower-limb joint biomechanics were computed from motion capture and 6-axis force plate data measured in the lab. Treadmill trials were completed using the inclinable, force instrumented treadmill. Overground trials were completed while participants walked over the modular force plates installed in the ramp, staircase, or level ground region depending on the trial. The overhead motion capture cameras measured participant kinematics within any region of the lab space based on retroreflective markers placed on the participant.

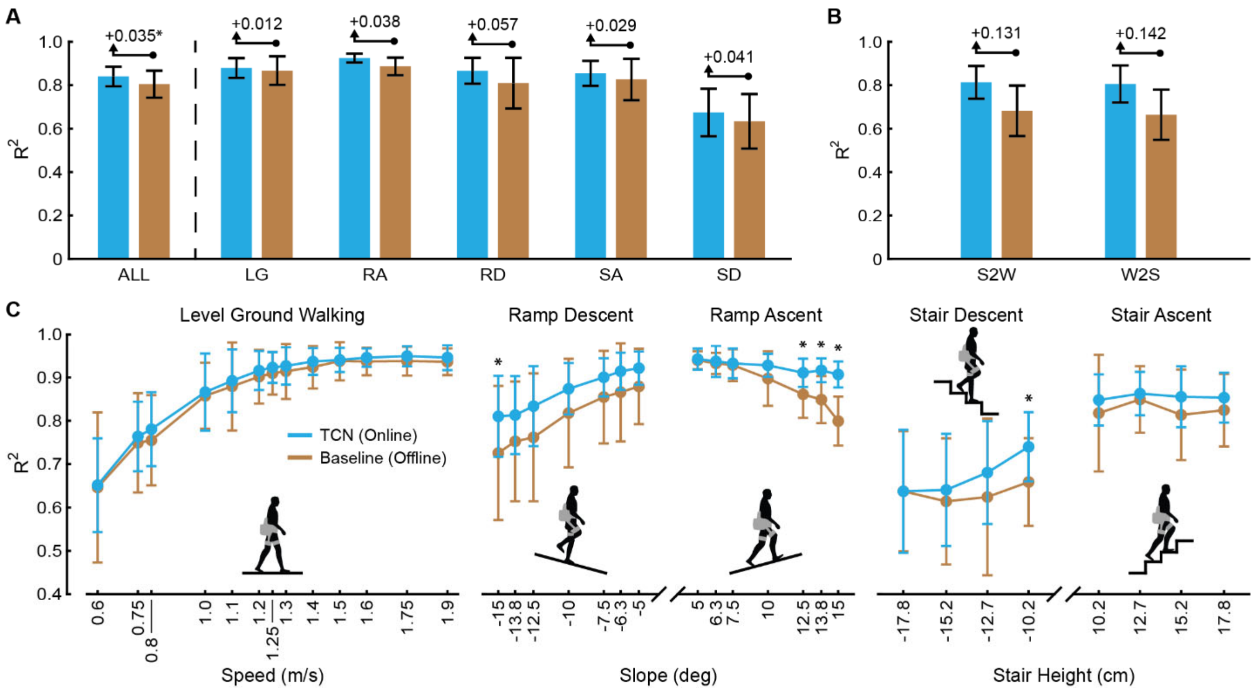

Fig. S6.

Hip moment estimation R^2. (**A**) The average R^2 of the temporal convolutional network (TCN) is compared to that of the Baseline method during level ground walking (LG), ramp ascent (RA), ramp descent (RD), stair ascent (SA), and stair descent (SD) walking (multiple comparisons, n = 10). The average R^2 across the five ambulation modes (ALL) is also shown (paired t-test, n = 10). (**B**) The average R^2 of the TCN is shown during stand-to-walk (S2W) and walk-to-stand (W2S) transitions relative to the Baseline method (n = 5, no statistical tests performed). (**C**) The average R^2 of the TCN and Baseline method is shown for each test condition within the LG, RA, RD, SA, and SD ambulation modes (multiple comparisons, n = 10 for all comparisons except for LG at 1.9 m/s and RD at -15°, which were n = 9). All TCN results are based on online estimates that were used in the control loop. All Baseline results were computed post hoc using the same data. Bars and markers represent means, error bars represent ±1 standard deviation about the mean, and asterisks indicate statistical significance ($P < 0.05$).

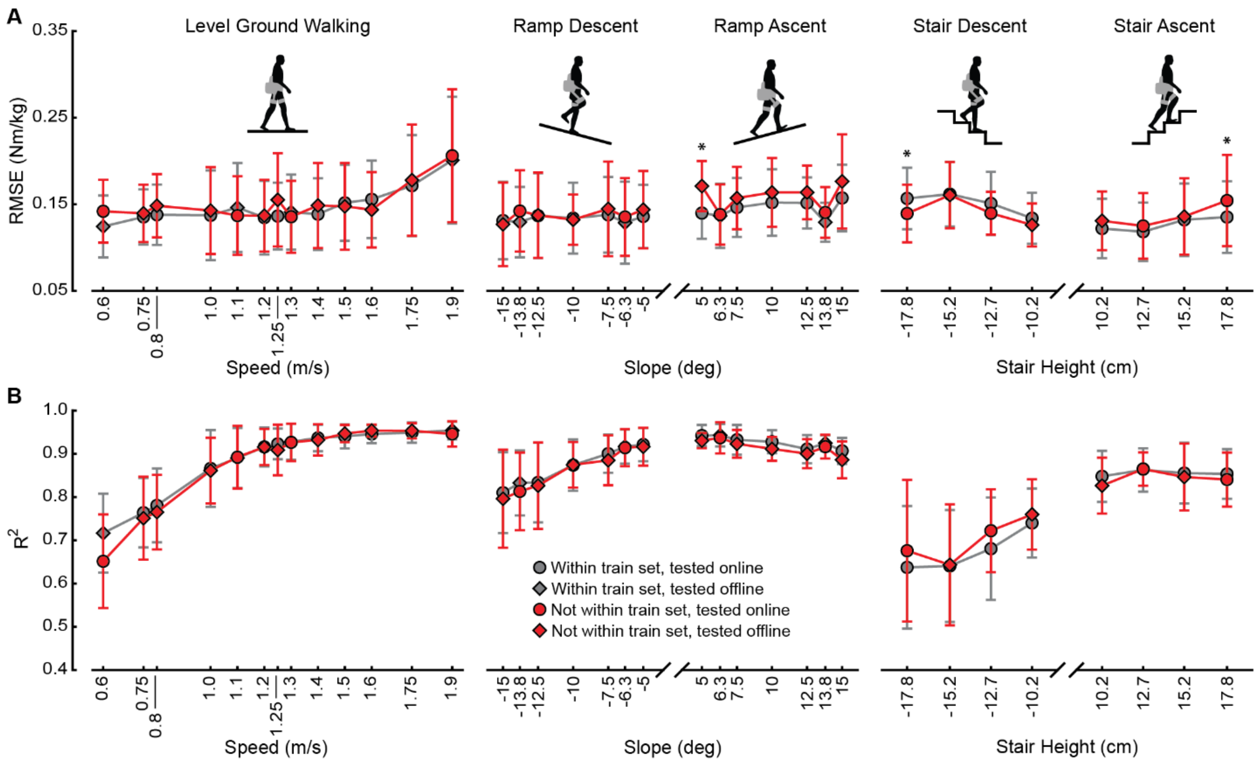

Fig. S7.

Model generalization to novel conditions. The temporal convolutional network (TCN) used to estimate user hip moments was tested across several ambulation modes and intensities. The TCN was tested on each ambulation mode and intensity when it was included in the training set (within train set) and was tested on each ambulation mode and intensity when it was not included in the training set (not within train set). Each condition was either evaluated online (when actively deployed on the hip exoskeleton and used in the control loop) or offline (tested post hoc on the same data). The results of the TCN under each condition are shown using (**A**) RMSE and (**B**) R^2. In each condition, the model was tested on data from participants that were not included in the training set (participant-independent evaluation). Markers represent means, error bars represent ±1 standard deviation about the mean, and asterisks indicate statistical significance (multiple comparisons, $P < 0.05$, $n = 10$ for all comparisons except for LG at 1.9 m/s, RD at -15°, and SA and SD at 12.7 cm, which were $n = 9$).

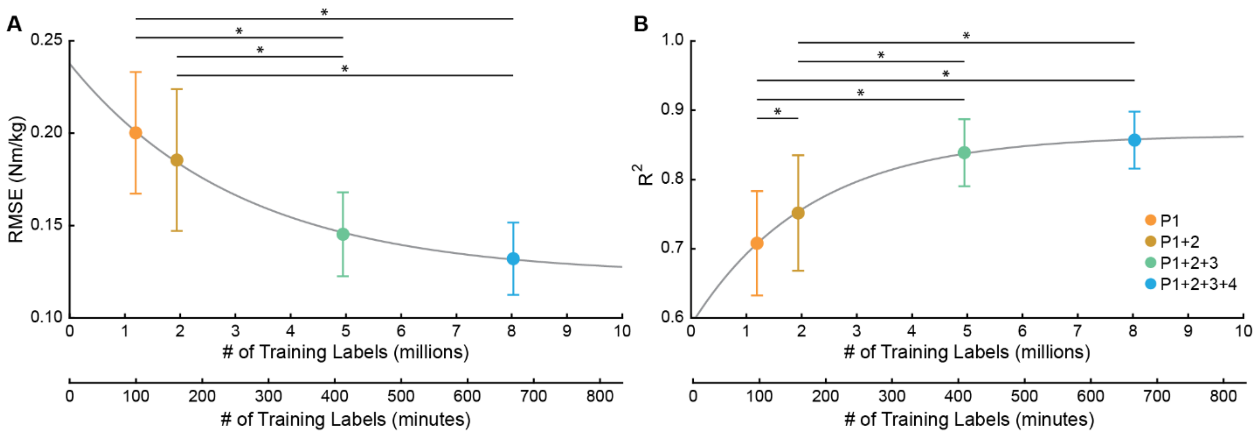

Fig. S8.

Effect of model training set size. The TCN was evaluated on the Phase 4 data when training the model using only the Phase 1 data (P1), using the Phase 1 and 2 data (P1+2), using the Phase 1, 2, and 3 data (P1+2+3), and when using all available data (P1+2+3+4). A leave-one-participant-out approach was used to train the TCN when both training and testing on the Phase 4 data (the P1+2+3+4 condition). (**A**) The resulting test RMSE when training the TCN under each condition shown. (**B**) The resulting test R^2 when training the TCN under each condition is shown. The number of training labels depicted for the P1+2+3+4 condition is the average across the leave-one-participant-out models since the horizontal error bar was too small to visualize. The x-axis is displayed both in total number of labels and in total number of minutes (training labels were collected at 200 Hz). Markers represent means, error bars represent ±1 standard deviation about the mean, and asterisks indicate statistical significance (multiple comparisons, $P < 0.05$, $n = 10$). The grey lines depict the trendline fit to the results using a decaying exponential function.

Table S1. Participant Information.
Relevant information of each participant in this study is provided.

Participant #	Gender	Height (m)	Body Mass (kg)	Age (years)
Experimental Phase 1				
AB01	Male	1.73	73.1	30
AB02	Female	1.75	73.8	20
AB03	Male	1.79	66.2	20
AB04	Female	1.67	62.3	23
AB05	Female	1.80	57.0	20
AB06	Male	1.82	84.0	22
AB07	Female	1.62	64.1	20
AB08	Male	1.76	73.4	21
AB09	Male	1.80	80.1	26
Mean ± SD	5 Males, 4 Females	1.75 ± 0.07	70.4 ± 8.7	22 ± 3
Experimental Phase 2				
AB10	Male	1.77	66.1	29
AB11	Male	1.83	71.2	23
AB12	Male	1.68	64.1	21
AB13	Male	1.70	69.6	28
AB14	Female	1.67	64.0	23
Mean ± SD	4 Males, 1 Female	1.73 ± 0.07	67.0 ± 3.3	25 ± 3
Experimental Phase 3				
AB15	Female	1.66	51.2	24
AB16	Male	1.74	73.8	22
AB17	Male	1.75	80.1	20
AB18	Male	1.85	90.8	25
AB19	Male	1.75	78.6	23
AB20	Male	1.83	90.5	25
AB21	Female	1.60	65.0	22
AB22	Male	1.74	60.7	31
AB23	Female	1.73	70.9	24
AB24	Male	1.80	78.6	20
Mean ± SD	7 Males, 3 Females	1.75 ± 0.07	74.0 ± 12.6	24 ± 3
Experimental Phase 4				
AB25	Male	1.83	92.9	18
AB26	Male	1.7	71.8	19
AB27	Male	1.7	82.5	26
AB28	Male	1.7	55.7	22
AB29	Male	1.63	66.2	25
AB30	Female	1.57	63.9	31
AB31	Male	1.83	87.5	30
AB32	Female	1.77	67.7	22
AB33	Male	1.65	61.2	34
AB34	Male	1.75	83.4	26
Mean ± SD	8 Males, 2 Females	1.71 ± 0.08	73.3 ± 12.5	25 ± 5

Table S2. Specifications of the Custom-Designed Robotic Hip Exoskeleton.

Relevant design specifications of the hip exoskeleton are provided. Peak torque was rated based on the maximum capability of the actuator and the requirement that the structural interface of the exoskeleton could provide this assistance to the user with minimal play between the user and device.

Parameter	**Value**
Peak Torque	18 Nm
Max. Continuous Torque	9 Nm
Max. Speed at Peak Torque	35.6 rad/s
Actuator Mass	0.49 kg
Total Exoskeleton Mass	4.8 kg
Hip Extension Range of Motion	-127° to 37°

Table S3. Optimization Constraints for Transforming Exoskeleton Sensor Data.
The upper and lower bounds of each optimization variable used to transform the inertial measurement unit data are provided.

Variable	Lower Bound	Upper Bound
$\hat{\theta}$	$-\pi$ rad	π rad
$\hat{\phi}$	$-\pi$ rad	π rad
$\hat{\psi}$	$-\pi$ rad	π rad
$\hat{p}_x$	-1 m	1 m
$\hat{p}_y$	-1 m	1 m
$\hat{p}_z$	-1 m	1 m

Data S1. Processed Study Data.

The processed data used to create all results of this study are provided as additional supplemental materials.

Data S2. Statistical Results.

The P-values and number of participants (n) from all statistical tests conducted in this study are provided.